

Causal Inference: Regression Discontinuity Designs

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Part 6

Overview

Strategies for estimating effects of treatments so far:

- Randomize treatment and take the DIGM
- Identify and control for confounding variables such that the CIA holds
- Identify a plausible counterfactual using time and unit variation in panel data
- Identify an instrumental variable and use two-stage-least-squares to estimate average treatment effect for compliers.

Today: take advantage of special situations where treatment is applied based on a cutoff

Discontinuities

“Natura non facit saltus.” (Nature makes no jump.)

- GOTTFRIED LEIBNIZ

But humans sometimes use rules that create jumps:

- Students pass with a 4.0 or better
- French municipalities use PR elections if population is 3,500 (now 1,000) or higher
- A candidate is elected if she receives more votes than any other candidate
- Journalists report a recession if the economy shrinks for two consecutive quarters

Regression Discontinuity Design (RDD)

- RDD is a fairly old idea (Thistlethwaite and Campbell, 1960) but this design experienced a renaissance in recent years
- Assignment to treatment and control is not random, but we know the **assignment rule** influencing how people are assigned or selected in to treatment
- Widely applicable in a rule based world (administrative programs, elections, etc.)
- High internal validity: In their validation study Cook and Wong (2008) identify RDD as one of the few observational study designs that can accurately reproduce experimental benchmarks

Assignment Mechanism

- Trying to do causal inference by comparing all units does not provide valid comparisons
- Often, better strategy is to study the assignment mechanism:

Definition (Assignment Mechanism)

Assignment mechanism is the procedure that determines which units get which treatment. Examples include:

- random assignment - sometimes not doable
 - selection on observables - usually unknown
 - selection on unobservables
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- Most statistical models (in causal inference) attain identification of treatment effects by restricting the assignment mechanism
 - RD uses assignment based on rules

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Hypothetical Illustration Example: Sharp RDD

- Thistlethwaite and Campbell (1960) study the effects of college scholarships on later students' achievements
- Scholarships are given on the basis of whether or not a student's test score exceeds some *arbitrary* cut-off c
 - Treatment D is scholarship
 - Forcing variable X is SAT score with cutoff c
 - Outcome Y is subsequent earnings
 - Y_0 denotes potential earnings without the scholarship
 - Y_1 denotes potential earnings with the scholarship
- Y_1 and Y_0 are correlated with X : on average, students with higher SAT scores obtain higher earnings

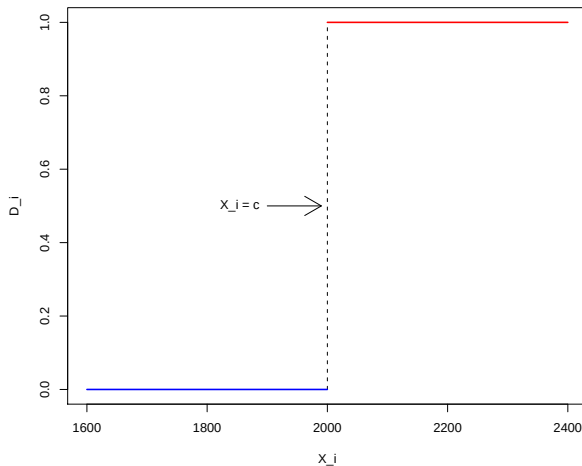
Sharp Regression Discontinuity Design

- Imagine a binary treatment D that is completely determined by the value of a predictor X_i being on either side of a fixed cutoff point c :

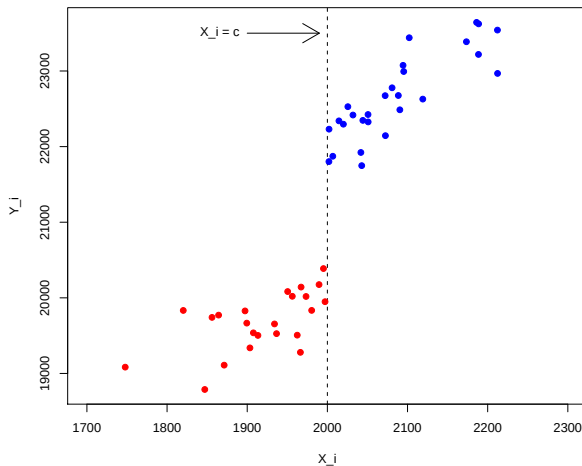
$$D_i = 1\{X_i > c\} \text{ so } D_i = \begin{cases} D_i = 1 & \text{if } X_i > c \\ D_i = 0 & \text{if } X_i < c \end{cases}$$

- X_i , called the **forcing variable**, may be correlated with the outcomes Y so comparing treated and untreated units does not provide causal estimates
 - (e.g. students with higher SAT scores obtain higher earnings even without the scholarship)
- But: We can use the discontinuity to find **as-good as random** variation

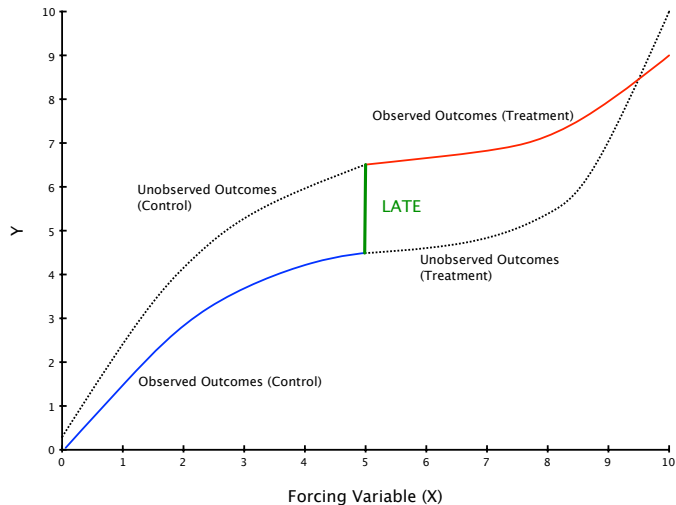
Sharp RDD: Graphical Illustration



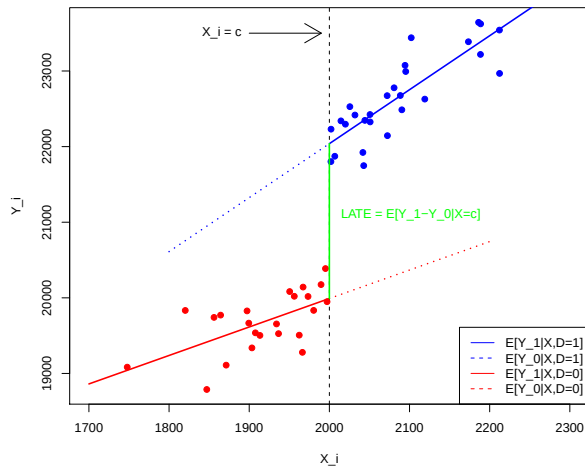
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Potential Outcomes

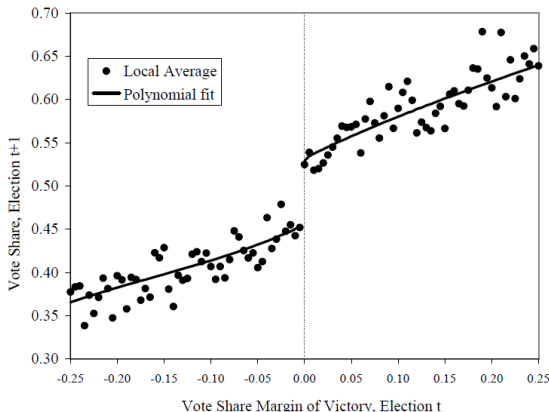


Sharp RDD: Graphical Illustration



Sharp RDD: Incumbency Advantage

Figure IVa: Democrat Party's Vote Share in Election $t+1$, by Margin of Victory in Election t : local averages and parametric fit



What about Jill Stein?



The New York Times

Jill Stein Won't Stop. No Matter Who Asks.

People in Stein's life have implored her to abandon her bid for president, lest she throw the election to Donald Trump. She's on the ballot in almost every critical state.



Outline

Introduction

Identification

Sharp RDD: Estimation

Example

Robustness

Review

Estimate Effect at Discontinuity

1. Trim the sample to a reasonable window around the threshold c (discontinuity sample):
 - $c - h \leq X_i \leq c + h$, where h is some positive value that determines the size of the window
2. Recode running variable to deviations from threshold: $\tilde{X} = X - c$.
 - $\tilde{X} = 0$ if $X = c$
 - $\tilde{X} > 0$ if $X > c$ and thus $D = 1$
 - $\tilde{X} < 0$ if $X < c$ and thus $D = 0$
3. Decide on a model for estimation, can be linear, quadratic, and so on

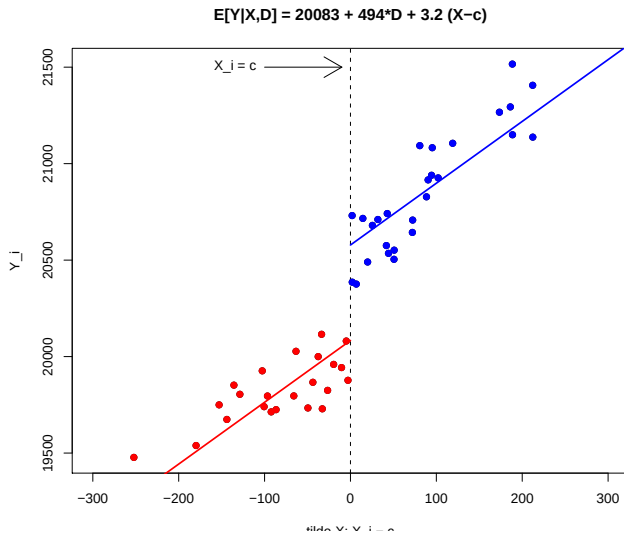
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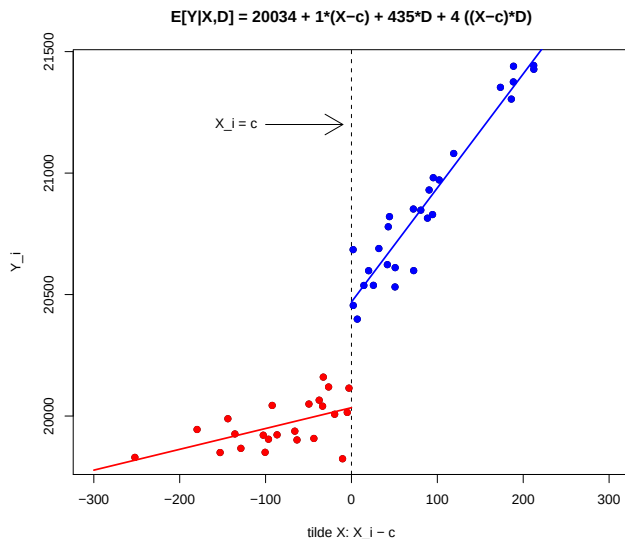
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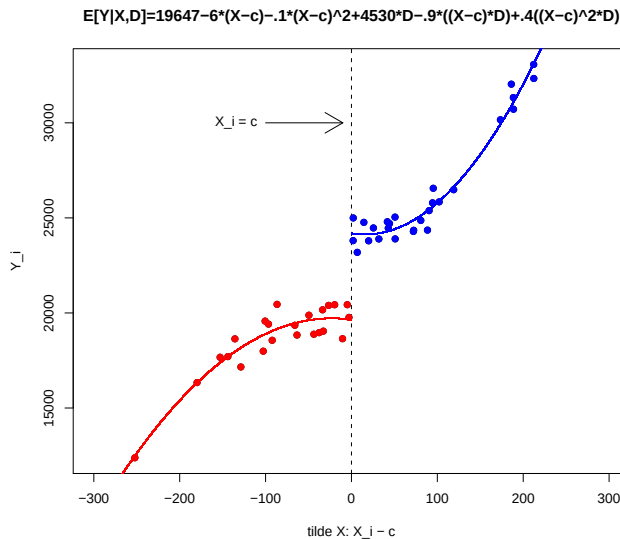
Linear with Same Slope

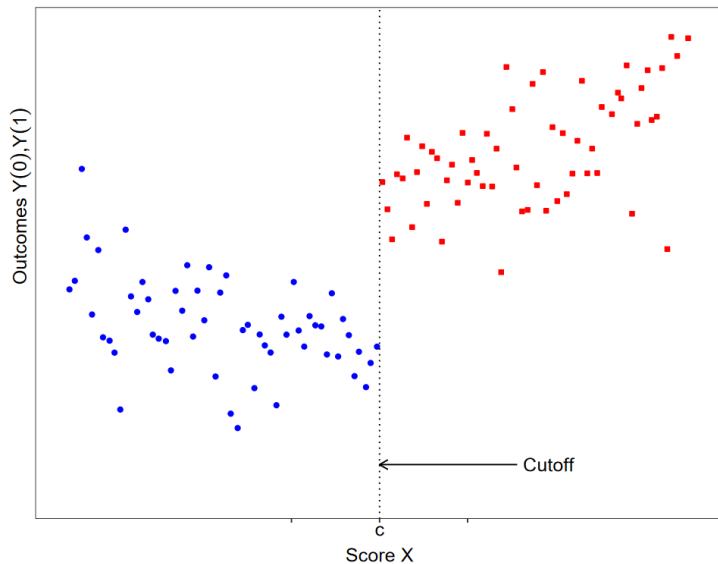


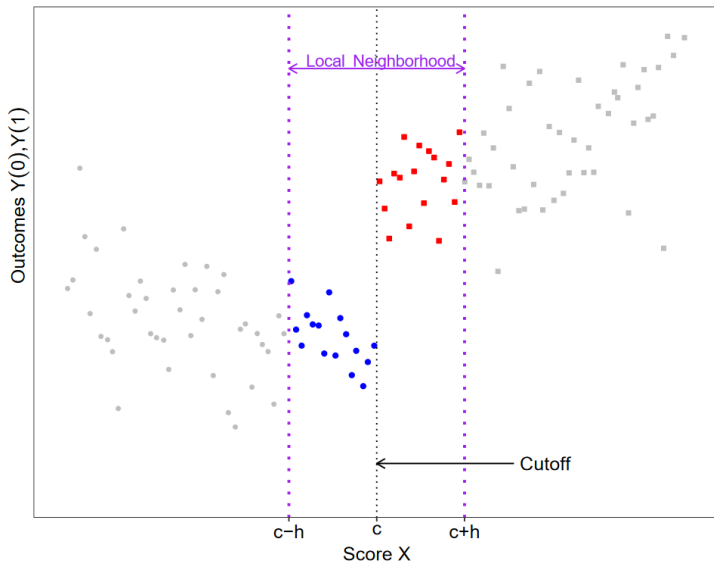
SRDD: Linear with Differential Slope

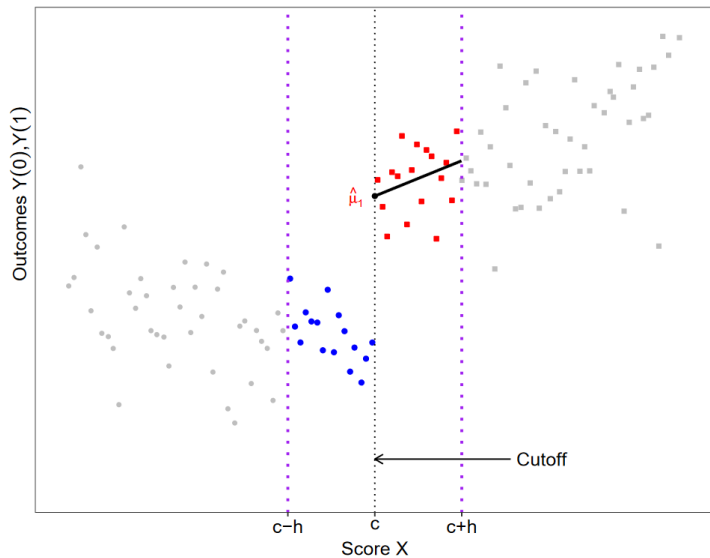


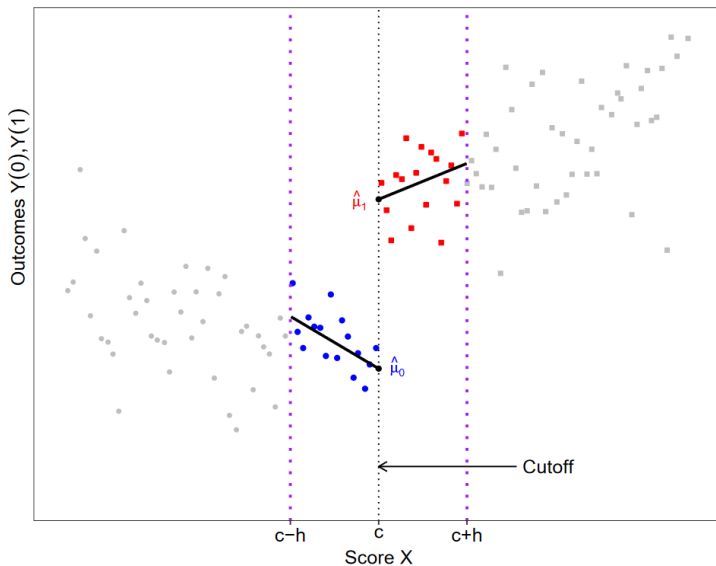
SRDD: Non-Linear Case











Which window size h should we use?

- How do we pick the bandwidth size?
- There is a trade-off between bias and efficiency:
 - Closer to the cut-off there is less bias in the estimator, but the sample size is smaller (= larger variance)
 - However, farther from the cut-off there is a larger sample size but there is a larger bias in the estimator

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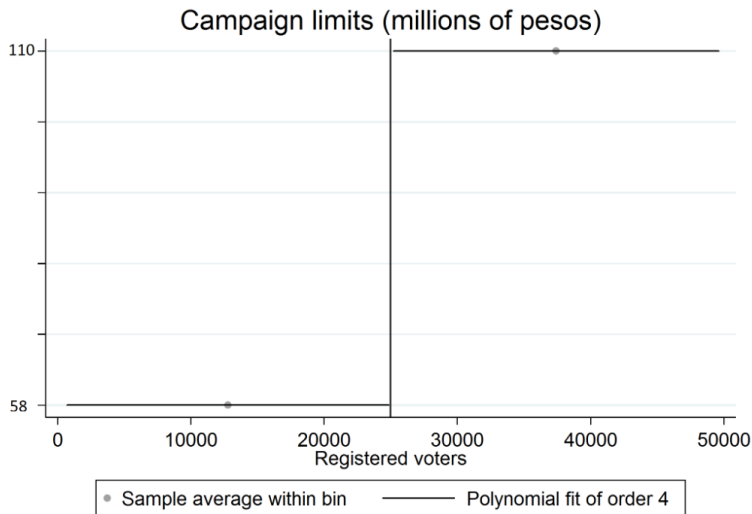
Example: Campaign Limits. Gulzar, Rueda and Ruiz (2020)

- Policymakers have enacted limits on private funding of campaigns to reduce the influence of donors
 - Over 40% of countries have some form of limit on how much donors are allowed to contribute to electoral campaigns IDEA (2014)
- But are they effective? What are the effects of campaign finance limits?
 - This paper: Effects of *campaign limits* on *public contracts*
 - They use a novel dataset from Colombia that links individual donations to public contracts and exploit institutional rules where campaign limits jump *discontinuously* at specific number of registered voters

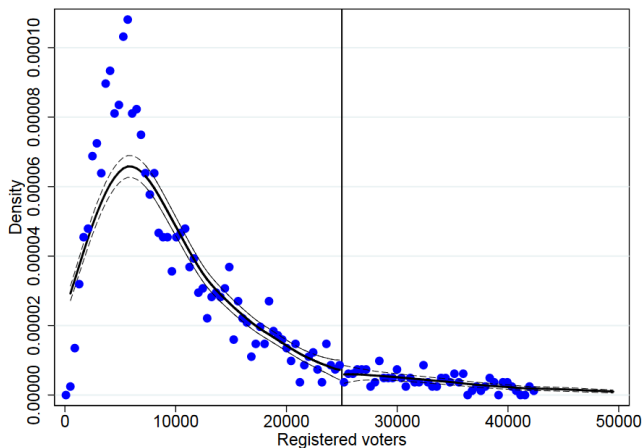
Context

- 75% of donors give contributions that are *larger* than the municipality average monthly wage
- Out of 6,658 donors in the dataset, 6,542 contribute only in one municipality
- There are **campaign finance limits** on total campaign contributions and on *individual contributions* (10% of total)
- Limits jump *discontinuously* at **arbitrary** registered voter cut-offs
- For example at 25,000 registered voters the campaign limit increases from 58M (17,000 USD) to 110M COP (36,400 USD)

Sharp Discontinuity



RDD-Distribution of registered voters



Notes: Figure shows the density of the running variable on both sides of the threshold, binned averages and 95% confidence intervals. The discontinuity

Empirical approach: RDD campaign limits

- Focus on similar municipalities where there are tighter **or** looser campaign limits by a narrow registered voter margin
- V_i = number of registered voters centered around 0 (25,000 cut-off)
- When $CampaignLimit_i = 1$ then ($V_i > 0$)

$$CampaignLimit_i = \begin{cases} CampaignLimit_i = 1 & \text{if } V_i > 0 \\ CampaignLimit_i = 0 & \text{if } V_i < 0 \end{cases} \quad (1)$$

- Note that the treatment implies **looser campaign limits**

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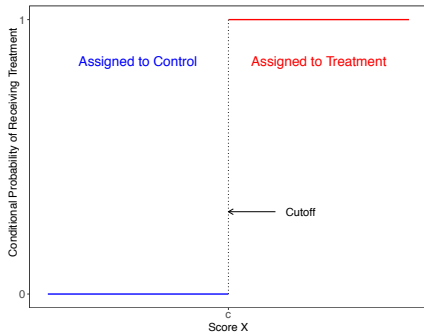
Looser limits and rewards for donors

Table: Effect of looser campaign contribution limits on contracts assigned to donors to the mayor

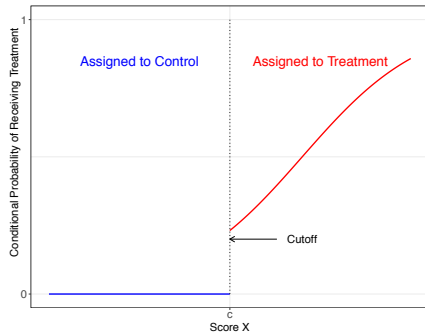
Outcome:	# Contracts (1)	ln(Value All) (2)	# Min. Value Contracts (3)	ln(Value Min. Value) (4)
Looser contribution limit	3.091	0.819	2.030	0.638
Robust p-value	0.012	0.234	0.023	0.061
CI 95%	[0.735,5.994]	[-0.513,2.098]	[0.306,4.200]	[-0.030,1.364]
Observations	2,049	2,049	2,049	2,049
Bandwidth obs.	457	366	366	341
Mean	0.280	0.205	0.210	0.101
Effect mean(%)	1,103.93	399.51	966.67	631.68
Bandwidth	6,980	5,312	5,292	5,190

Local linear estimates of average treatment effects at cutoff estimated with triangular kernel weights and optimal MSE bandwidth. 95% robust confidence intervals and robust p-values with clustering at the municipality level are computed following Calonico, Cattaneo and Titiunik (2014). Bandwidth obs. denotes number of observations in the optimal MSE bandwidth for each dependent variable. Each observation is a donor.

Sharp vs Fuzzy RDD



(a) Sharp RD



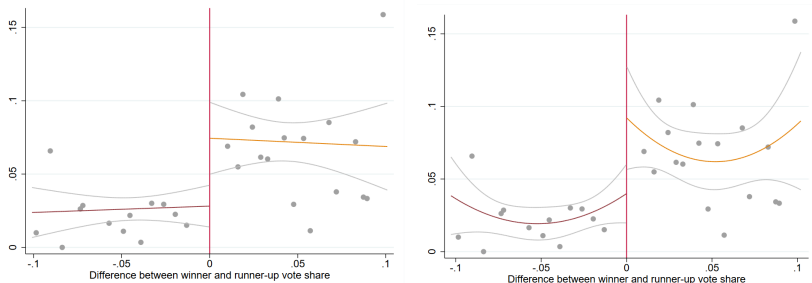
(b) Fuzzy RD (one-sided compliance)

RDD: Falsification Checks

1. Sensitivity: Are results sensitive to alternative specifications?
2. Continuity Checks: Do predetermined covariates X jump at the threshold?
 - Do placebo outcomes jump at the threshold?
3. Sorting: Do units sort around the threshold?
4. Check if jumps occur at placebo thresholds c^* (?)

Example of graphical representation of result

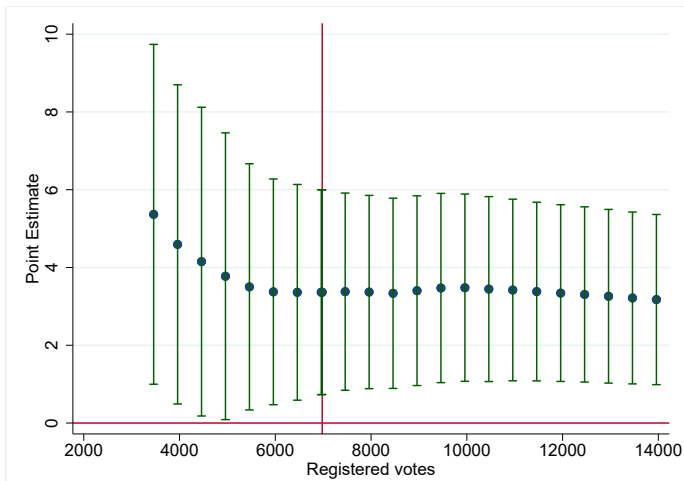
Effect of barely electing a politician on probability of donor getting a contract



Note: Observations within Calonico et. al (2014) bandwidth displayed. Left: linear fit. Right: quadratic fit.

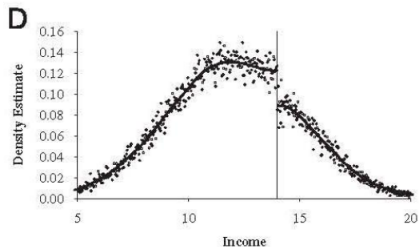
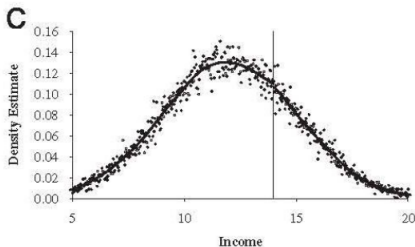
Estimate for different bandwidth sizes

Effect of looser campaign limits on number of contracts for donors



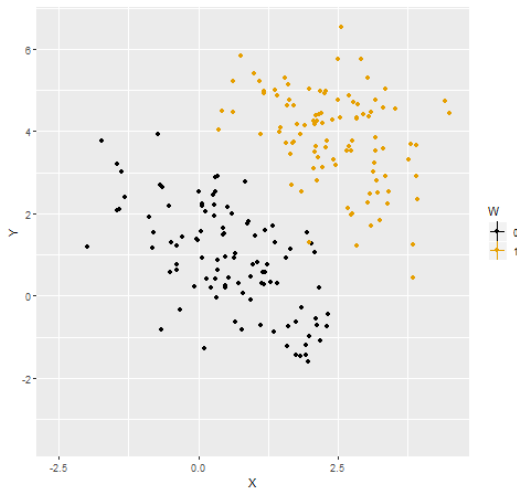
RDD: Sorting Around the Threshold

Example: Beneficial job training program offered to agents with income $< c$. Concern, people will withhold labor to lower their income below the cut-off to gain access to the program.



Variation - Conditioning

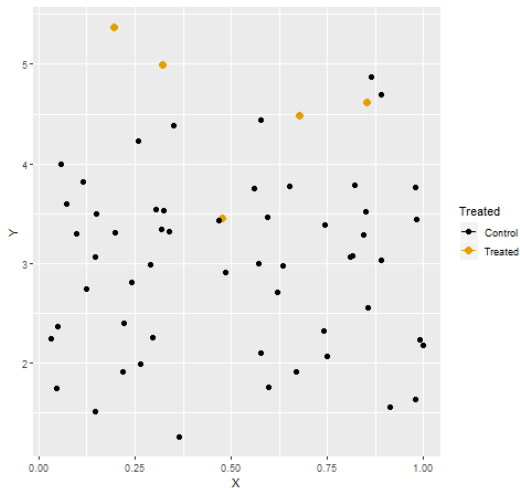
The Relationship between Y and X, Controlling for a Binary Variable W
1. Start with raw data. Correlation between X and Y: 0.319



Variation - Matching

The Effect of Treatment on Y while Matching on X (with a caliper)

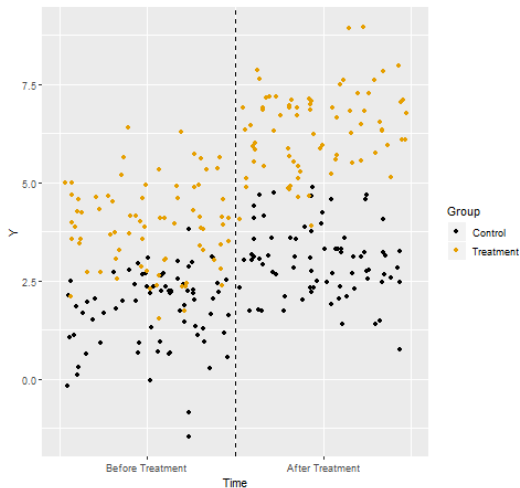
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Variation - DiD

The Difference-in-Difference Effect of Treatment

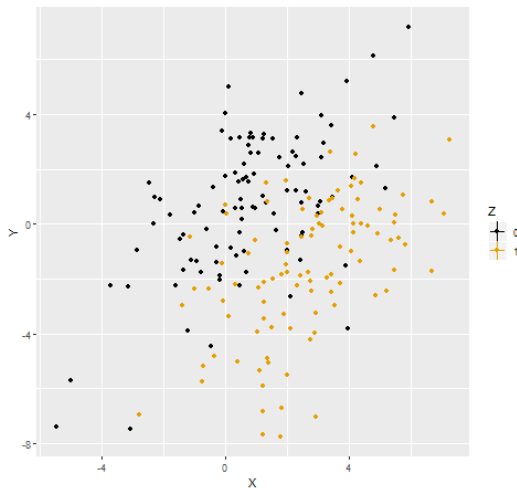
1. Start with raw data.



Variation - IV

The Relationship between Y and X, With Binary Z as an Instrumental Variable

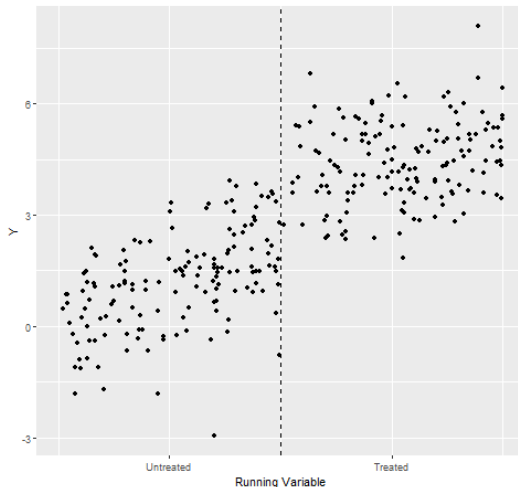
1. Start with raw data. Correlation between X and Y: 0.282



Variation - RDD

The Effect of Treatment on Y using Regression Discontinuity

1. Start with raw data.



Concluding Remarks

- Context matters! Think about counterfactuals and whether you really capture the treatment effect.
- What about your own research?
- Get in touch!
- Thank you!